**Practical No. 2**

Aim - Use Telegram to control Raspberry Pi GPIO.

Apparatus – Breadboard, LED, Resister, Raspberry Pi 4, Jumper wires.

Install telegram desktop on the Rasberry pi -

1. **For idle installation sudo apt-get install idle**
2. **Install Telegram Bot on Raspberry pi sudo pip install telepot**

Create a Telegram Bot
Log in using your phone number and the verification code sent to your Telegram account.
Use Telegram on your phone
Search - **BotFather**
Create a bot. Send
**/newbot**
Example -
BotFather: Please choose a name.
You: rpi4control_ab
BotFather: Choose a username ending with bot.
You: **rpi4control_abbot**
BotFather returns : Done! Congratulations …….
Token to access the HTTP API :
**892375566778787972:AAFuICyBNTmbchjhgXR0oe4OQaw4tfPtYe-I**
Save this. This is your Bot Token.

1. **Connect 4 LED with Raspberry Pi's GPIO Pins.**

| Led Terminal | Pin Number | GPIO Number |
|---|---|---|
| Green Positive | Pin 3 | GPIO 2 |
| Yellow Positive | Pin 5 | GPIO 3 |
| Red Positive | Pin 7 | GPIO 4 |
| Blue Positive | Pin 11 | GPIO 17 |
| Negative of all four | Pin 9 | GND |

2. **Install Telegram App in Mobile. Follow process to obtain access token**

```
Raspberry Pi Telegram Bot
Install the Telegram app on your smartphone from the Play Store.
Open Telegram.
Request BotFather to create a new bot.
Search for @BotFather and click Start.
Create a new bot using the command:
/newbot
Provide a name for your bot (e.g., practyit).
Then provide a username for your bot.
The username must end with bot (e.g., prac_bot).
After completing the process, BotFather will provide you with a bot token for access.
This token is what you'll use in your Raspberry Pi program to authenticate and control
your Telegram bot. Keep it private, as anyone with the token can control your bot.
```

### 3. Write Python Script to blink LED with Telegram Bot

```
import time
import datetime
import RPi.GPIO as GPIO
import telepot
from telepot.loop import MessageLoop

# GPIO pin numbers
green = 2
yellow = 3
red = 4
blue = 17

now = datetime.datetime.now()

GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

# LED Setup
GPIO.setup(blue, GPIO.OUT)
GPIO.output(blue, 0)

GPIO.setup(yellow, GPIO.OUT)
GPIO.output(yellow, 0)

GPIO.setup(red, GPIO.OUT)
GPIO.output(red, 0)

GPIO.setup(green, GPIO.OUT)
GPIO.output(green, 0)


def action(msg):
    global chat_id

    chat_id = msg['chat']['id']
    command = msg['text'].lower()

    print("Received: {}".format(command))

    if "on" in command:
        message = "On "

        if "blue" in command:
            message += "blue "
            GPIO.output(blue, GPIO.HIGH)
```

```
    if "yellow" in command:
        message += "yellow "
        GPIO.output(yellow, GPIO.HIGH)

    if "red" in command:
        message += "red "
        GPIO.output(red, GPIO.HIGH)

    if "green" in command:
        message += "green "
        GPIO.output(green, GPIO.HIGH)

    if "all" in command:
        message += "all "
        GPIO.output(blue, GPIO.HIGH)
        GPIO.output(yellow, GPIO.HIGH)
        GPIO.output(red, GPIO.HIGH)
        GPIO.output(green, GPIO.HIGH)

    message += "light(s)"
    telegram_bot.sendMessage(chat_id, message)

elif "off" in command:
    message = "Off "

    if "blue" in command:
        message += "blue "
        GPIO.output(blue, GPIO.LOW)

    if "yellow" in command:
        message += "yellow "
        GPIO.output(yellow, GPIO.LOW)

    if "red" in command:
        message += "red "
        GPIO.output(red, GPIO.LOW)
```

```
        if "green" in command:
            message += "green "
            GPIO.output(green, GPIO.LOW)

        if "all" in command:
            message += "all "
            GPIO.output(blue, GPIO.LOW)
            GPIO.output(yellow, GPIO.LOW)
            GPIO.output(red, GPIO.LOW)
            GPIO.output(green, GPIO.LOW)

        message += "light(s)"
        telegram_bot.sendMessage(chat_id, message)


# Replace with your Telegram Bot Token
telegram_bot = telepot.Bot("YOUR TOKEN")

print(telegram_bot.getMe())

MessageLoop(telegram_bot, action).run_as_thread()

print("Up and Running...")

while True:
    time.sleep(10)
```

**To Run the Program Eg. python test.py**

Commands & O/P -

| User sends - | Result |
|---|---|
| /start | Shows the list of available commands |
| /blue | LED ON |
| /red | LED ON |
| /all | LED ON |

